# KRITICAL HACKS!

## TEAM NAME: BITHEADS

**PROBLEM STATEMENT:**
Students risk losing years and money to unrecognised colleges and invalid degrees.

**VERIXA AI:**
An AI-powered web agent that tells students, in  few seconds , whether a college and course is legally recognised for the admission year.

**TEAM MEMBERS NAME:**
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## What makes college or course recognition difficult to verify?

- **Unrecognised admission**
  not recognised or not approved for that year
- **Late discovery**
  only after paying fees, sometimes years later
- **Fragmented verification**
  separate portals and lists
- **Year-specific approval**
  may lose approval in another year
- **Inconsistent names**
  many spellings and abbreviations across sources
- **Biased sources**
  financial interest in the sale

# PROBLEM STATEMENT

Every year, lakhs of students take admission in colleges or courses that are not recognised, not approved for that year, or falsely claim affiliation.

- Students find out only after paying fees, sometimes years later, when the degree is rejected for jobs, higher studies or licensing exams.
- Verification is fragmented. UGC, AICTE, NMC, BCI, PCI and state universities each keep separate portals and lists.
- Approval is year-specific. A college approved in one year may lose approval in another.
- Names are inconsistent. One institution appears under many spellings and abbreviations across sources.
- Students rely on biased sources. Brochures and admission agents have a financial interest in the sale.

# PROPOSED SOLUTION

*An AI-powered web agent system that checks a college's legitimacy live, across all relevant regulators, in few seconds .*

*The student enters the college name, course and admission year. The system then:*

| | |
|---|---|
| _01 | IDENTIFIES WHICH REGULATORS APPLY TO THAT COURSE. |
| _02 | SENDS WEB AGENTS TO EACH REGULATOR'S LIVE PORTAL. |
| _03 | MATCHES THE INSTITUTION ACROSS SOURCES DESPITE NAME VARIATIONS. |
| _04 | EXTRACTS STATUS, APPROVAL YEAR, COURSE AND AFFILIATION. |
| _05 | RETURNS A CLEAR VERDICT WITH EVIDENCE |
| _06 | SHOWS A PLAIN-LANGUAGE EXPLANATION, SOURCE LINKS AND A "WHAT TO DO NEXT" CHECKLIST. |

# TARGET USERS AND USE CASES

**1. Students and 12th-Pass Aspirants**
Students can use the platform to verify a college's legal recognition and check whether it is authorised to offer their chosen course before paying admission fees.

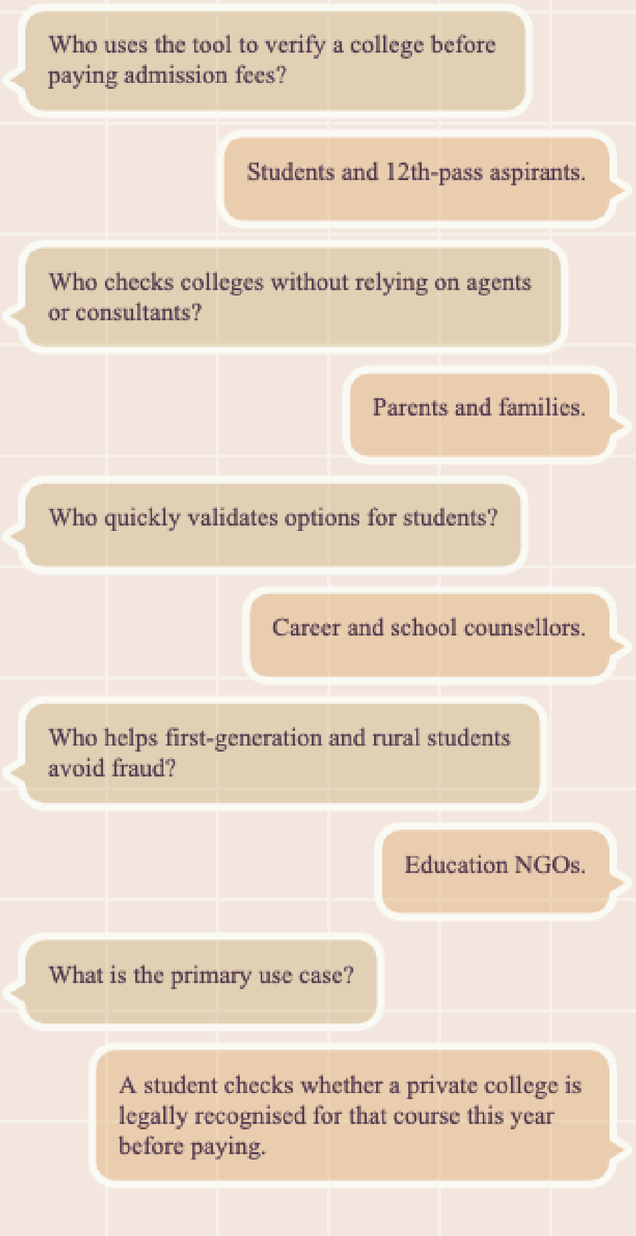
**2. Parents and Families**
Parents can independently check the legitimacy of colleges without relying solely on admission agents, consultants, or promotional claims. .

**3. Career and School Counsellors**
Career counsellors and school advisors can quickly verify colleges and courses while guiding students towards legitimate educational opportunities.

**4. Education NGOs**
Education NGOs working with first-generation learners and students from rural or underserved communities can use the platform to identify potentially fraudulent institutions.

**5. Employers and HR Teams (Secondary Users)**
Employers and HR professionals can verify whether a candidate's educational institution was officially recognised during the year they graduated.

**Primary Use Case:** *A student receives an admission offer from a private college and is asked to pay an admission fee. Before making the payment, the student uses the platform.*



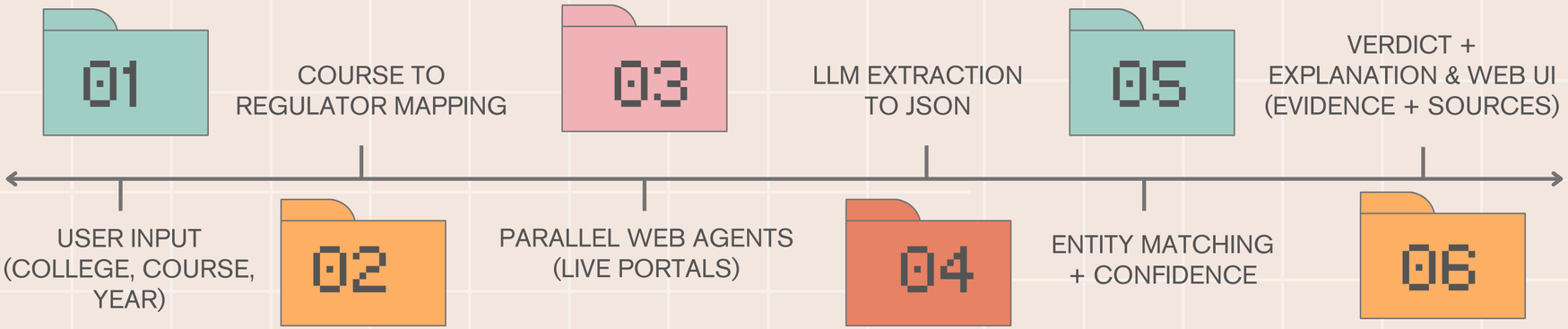
# PROPOSED TECHNICAL APPROACH

- **FRONTEND AND BACKEND**
  - Vite + TypeScript, Three.js
  - Python + FastAP, SQLAlchemy
- **WEB AGENTS& DATA ACQUISION, STORAGE & DATA MANAGEMENT**
  - TinyFish Agent API, TinyFish Search API, TinyFish Fetch API, Playwright + BeautifulSoup
  - SQLite, SQLAlchemy Models
- **ML&LLM**
  - Fuzzy Matching, Embeddings, LLM-based Extraction
  - Confidence Scoring, Verdict Engine, LLM Explanation Generation

# ROLE OF ML AND AI

| COMPONENT | WHAT IT DOES |
|---|---|
| 1. AI-Based Data Extraction | Official websites aur regulatory documents se relevant information extract karna. |
| 2. Entity Resolution | Different formats mein likhe institute names ko match karna. |
| 3. Confidence Scoring | Source reliability, evidence consistency aur data recency ke basis par confidence estimate karna. |
| 4. Intelligent Verdict Engine | Available evidence aur verification rules ke basis par institution ka status classify karna. |
| 5. AI-Generated Explanation | Verdict ke saath supporting evidence aur limitations ko simple language mein explain karna. |

# OUR PIPELINE




## PROPOSED TINYFISH INTEGRATION

***Why TinyFish is essential:*** *Regulator portals are mostly search forms and dynamic tables with no clean API. A web agent has to type, filter, paginate and read results the way a human would.*

| TINYFISH CAPABILITY | HOW WE USE IT |
|---|---|
| 1. Agent API | Navigates regulator portals, enters the college name, applies filters and extracts result tables |
| 2. Parallel execution | Checks UGC, AICTE and the course-specific body at the same time, so results return fast |
| 3. Search API | Finds the official affiliating university page and any public notices about the institution |
| 4. Fetch API | Converts official PDFs and notice pages into clean text for extraction |

# EXPECTED IMPACT

### Level information gap

First-generation and small-town families get the same verification power

### Verification Burden

A check takes hours across multiple portals

### Save years

Avoid losing years of study that turn out to be invalid

### Prevent financial loss

Students avoid paying lakhs for invalid degrees

### Build trust

Every result comes with source links

### Reduce fraudulent reach

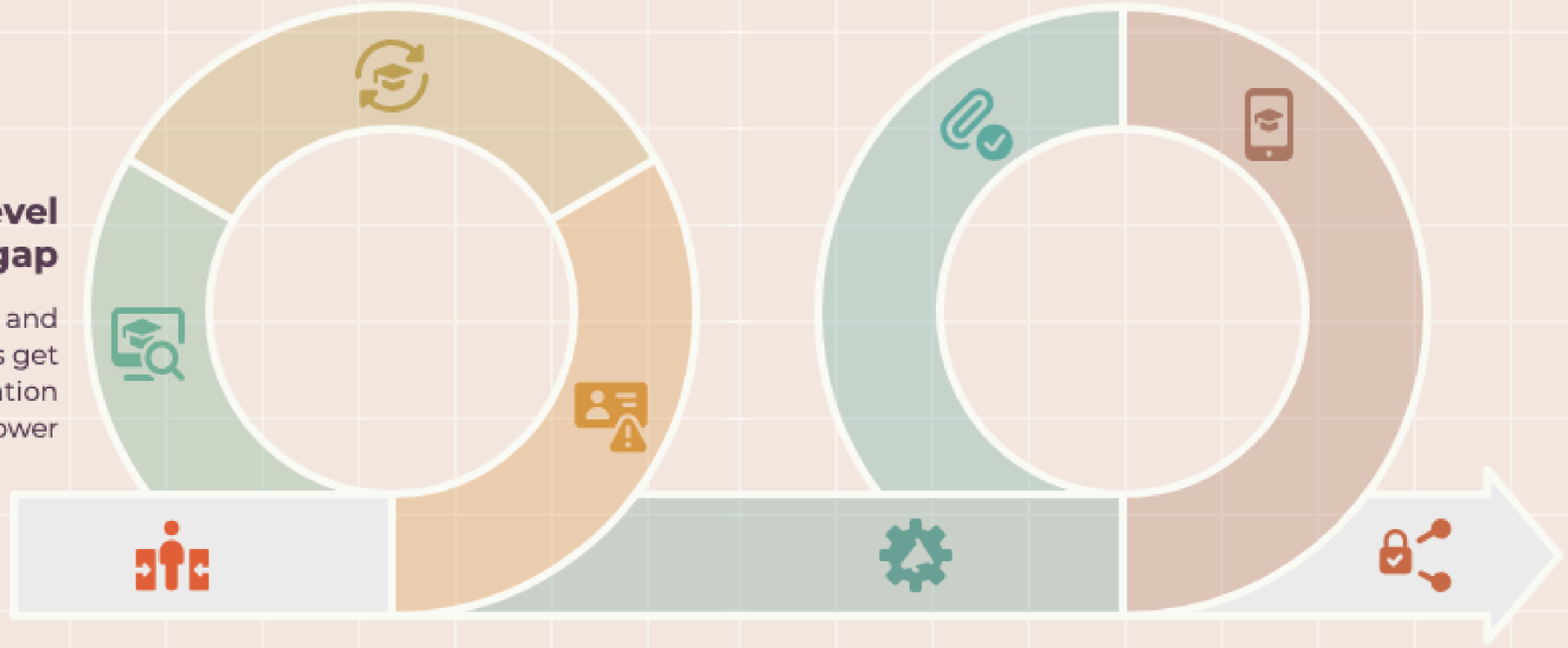
Verified information makes misleading admission claims harder to sell

### Save time

A check takes under a minute

### Trusted Results

Users can verify independently



# FEASIBILITY

## 01 TECHNICAL FEASIBILITY

- Uses public regulator data and existing tools (TinyFish API, standard ML libraries).
- A working prototype can be built in 24 hours with a narrow scope.
- Low cost: runs on API credits and free-tier hosting.

## 02 MVP SCOPE

- **Course**: B.Tech (AICTE and university affiliation), optionally one more
- **Region:** Delhi-NCR colleges
- **Regulators:** 2-3 portals tested in advance

## Risk Mitigation Strategies

| | Mitigation |
|---|---|
| **Portal blocks or captchas** | Test early; drop blocked portals from the MVP. |
| **Outdated or wrong data** | Show source link and date; advise confirming on the official portal. |
| **Name-matching errors** | Confidence score and a "Doubtful" category. |
| **Legal / defamation risk** | Neutral wording ("not found in official list") and always cite the source. |
| **Terms of use** | Public pages only; reasonable request rates. |

# FUTURE SCOPE

- Expand to all courses (medical, law, pharmacy, management, diplomas) and all states.
- Hindi and regional-language support.
- Browser extension that flags a college's status while you browse its website.
- WhatsApp bot so families can check by chat, without needing a new app.
- Alerts when a college's approval status changes.
- Employer verification for HR teams.
- Open API for counselling platforms and education portals.
- Partnerships with education NGOs and government awareness campaigns.